

Thème 3 — Fractions rationnelles • Corrigé Moyen

Corrigé 1

$\frac{2x}{(x+1)(x+3)} = \frac{A}{x+1} + \frac{B}{x+3}$. $A = \frac{2x}{x+3} \Big|_{x=-1} = \frac{-2}{2} = -1$, $B = \frac{2x}{x+1} \Big|_{x=-3} = \frac{-6}{-2} = 3$.
Contrôle : $-1(x+3) + 3(x+1) = -x-3+3x+3 = 2x$ ✓. Donc

$$\int \frac{2x}{(x+1)(x+3)} dx = -\ln|x+1| + 3\ln|x+3| + C.$$

Corrigé 2

$\frac{2x-1}{(x+2)(x-3)} = \frac{A}{x+2} + \frac{B}{x-3}$. $A = \frac{2x-1}{x-3} \Big|_{x=-2} = \frac{-5}{-5} = 1$, $B = \frac{2x-1}{x+2} \Big|_{x=3} = \frac{5}{5} = 1$.
Contrôle : $1(x-3) + 1(x+2) = 2x-1$ ✓. Donc

$$\int \frac{2x-1}{(x+2)(x-3)} dx = \ln|x+2| + \ln|x-3| + C.$$

Corrigé 3

Factorisation. $x^2 - x - 6$ a pour racines 3 et -2 (somme 1, produit -6), donc $x^2 - x - 6 = (x-3)(x+2)$.

$\frac{x+7}{(x-3)(x+2)} = \frac{A}{x-3} + \frac{B}{x+2}$. $A = \frac{x+7}{x+2} \Big|_{x=3} = \frac{10}{5} = 2$, $B = \frac{x+7}{x-3} \Big|_{x=-2} = \frac{5}{-5} = -1$.
Contrôle : $2(x+2) - 1(x-3) = 2x+4-x+3 = x+7$ ✓. Donc

$$\int \frac{x+7}{x^2-x-6} dx = 2\ln|x-3| - \ln|x+2| + C.$$

Corrigé 4

Factorisation. $x^2 - 4 = (x-2)(x+2)$.

$\frac{x}{(x-2)(x+2)} = \frac{A}{x-2} + \frac{B}{x+2}$. $A = \frac{x}{x+2} \Big|_{x=2} = \frac{2}{4} = \frac{1}{2}$, $B = \frac{x}{x-2} \Big|_{x=-2} = \frac{-2}{-4} = \frac{1}{2}$.
Contrôle : $\frac{1}{2}(x+2) + \frac{1}{2}(x-2) = x$ ✓. Donc

$$\int \frac{x}{x^2-4} dx = \frac{1}{2}\ln|x-2| + \frac{1}{2}\ln|x+2| + C = \frac{1}{2}\ln|x^2-4| + C.$$

Corrigé 5

Factorisation. $x^2 + x - 2$ a pour racines 1 et -2 (somme -1 , produit -2), donc $x^2 + x - 2 = (x-1)(x+2)$.

$\frac{3x+1}{(x-1)(x+2)} = \frac{A}{x-1} + \frac{B}{x+2}$. $A = \frac{3x+1}{x+2} \Big|_{x=1} = \frac{4}{3}$, $B = \frac{3x+1}{x-1} \Big|_{x=-2} = \frac{-5}{-3} = \frac{5}{3}$. *Contrôle* :
 $\frac{4}{3}(x+2) + \frac{5}{3}(x-1) = \frac{4}{3}x + \frac{8}{3} + \frac{5}{3}x - \frac{5}{3} = \frac{9}{3}x + \frac{3}{3} = 3x+1$ ✓. Donc

$$\int \frac{3x+1}{x^2+x-2} dx = \frac{4}{3}\ln|x-1| + \frac{5}{3}\ln|x+2| + C.$$